M378K Introduction to Mathematical Statistics
Problem Set #18
Consistency.

Definition 18.1. *$\hat{\theta}_n$ is said to be a* consistent estimator *of θ if*

$$\hat{\theta}_n \to \theta \quad \text{in probability as } n \to \infty,$$

i.e., if for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left[|\hat{\theta}_n - \theta| > \varepsilon\right] = 0.$$

$\mathbb{E}[\hat{\theta}_n]$

Markov Inequality
Chebyshev Inequality

Theorem 18.2. *Let $\hat{\theta}_n$ be* unbiased *and such that*

$$\mathrm{Var}\left[\hat{\theta}_n\right] \xrightarrow{n\to\infty} 0.$$

Then, $\hat{\theta}_n$ is a **consistent estimator**.

Problem 18.1. *Let $Y_1, Y_2, \ldots, Y_n$ be a random sample from any distribution with finite first and second moments. Propose a consistent estimator for the population mean μ and* **prove** *that it is, indeed, consistent.*

→: Propose: $\bar{Y}_n = \frac{1}{n}(Y_1 + \cdots + Y_n)$

Criteria for consistency:

- unbiased ? ✓
- $\mathrm{Var}[\bar{Y}_n] \xrightarrow{n} 0$

$$= \frac{\mathrm{Var}[Y_1]}{n} \quad (n < \infty)$$

$$= \frac{\mathbb{E}[Y_1^2] - (\mathbb{E}[Y_1])^2}{n} \quad (< \infty)$$

$\downarrow n$

0

□

Problem 18.2. *Consider a random sample $Y_1, Y_2, \ldots, Y_n$ from a power distribution with the density of the form*

$$f_Y(y) = \theta y^{\theta-1} \mathbf{1}_{(0,1)}(y).$$

*What is a consistent estimator for $\frac{\theta}{\theta+1}$? **Prove** that your choice is indeed consistent.*

$$\mathbb{E}[Y_1] = \int_0^1 y \cdot \theta \cdot y^{\theta-1} dy = \theta \int_0^1 y^\theta dy = \theta \cdot \frac{y^{\theta+1}}{\theta+1}\Big|_{y=0}^1$$

$$\boxed{\mathbb{E}[Y_1] = \frac{\theta}{\theta+1}} \checkmark$$

We propose: $\bar{Y}_n$

Our criteria:

- unbiased ✓
- $\mathrm{Var}[\bar{Y}_n] \xrightarrow[n\to\infty]{} 0$

If $\mathrm{Var}[Y_1] < \infty$, we're done

$$\mathrm{Var}[Y_1] = \mathbb{E}[Y_1^2] - (\mathbb{E}[Y_1])^2 < \infty \checkmark$$

$$\int_0^1 y^2 \cdot \theta \cdot y^{\theta-1} dy = \theta \int_0^1 y^{\theta+1} dy = \frac{\theta}{\theta+2} < \infty$$

Example. • $Y_1, \dots, Y_n$ random sample from $E(\tau = \theta)$

$\hat{\theta}_n = n \cdot Y_{(1)}$ unbiased for all n

$Y_{(1)} \sim E(\tau/n)$

Consider $\mathrm{Var}[\hat{\theta}_n] = \mathrm{Var}[n \cdot Y_{(1)}]$

$= n^2 \mathrm{Var}[Y_{(1)}] = n^2 \cdot \frac{\tau^2}{n^2} = \tau^2$

$Y_{(1)} \sim E(\tau/n)$

$\not\to n$

$\to 0$

The criterion is not satisfied.

• Task: Find an unbiased estimator for θ if

$Y_1, \dots, Y_n \sim U(0, \theta)$

Is it consistent?

Q: $\mathbb{E}[(Y-a)^2] \xrightarrow[a]{} \min$ $\quad a^* = \mathbb{E}[Y]$ ←

$\mathbb{E}[|Y-a|] \xrightarrow[a]{} \min$ $\quad a^* = \text{median of } Y$

Maximum Likelihood Estimation.

Likelihood.

Def'n. Given a random sample $Y_1, \dots, Y_n$ from a discrete distribution w/ an unknown parameter θ, the likelihood function is defined as

$$L(\theta; y_1, \dots, y_n) = p^{\theta}_{Y_1, \dots, Y_n}(y_1, \dots, y_n)$$
$$= p^{\theta}_{Y_1}(y_1) \cdot p^{\theta}_{Y_2}(y_2) \cdot \dots \cdot p^{\theta}_{Y_n}(y_n)$$
$$= p^{\theta}(y_1) \cdot p^{\theta}(y_2) \cdot \dots \cdot p^{\theta}(y_n)$$

where p^{θ} is the pmf of $\mathcal{D}$.